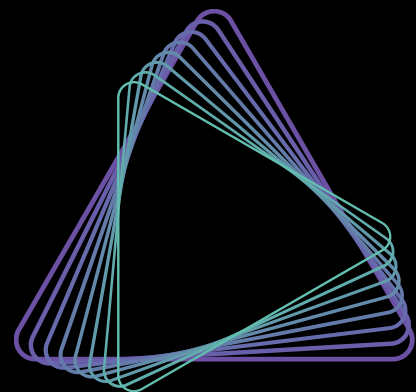




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基因数据库注释

Chunmei.Tao

2024/7

分析内容展示——单细胞转录组



10X Genomics单细胞转录组数据分析流程

简介

AnimalTFDB v4.0



基因数据库注释是指参考相关数据库对基因的功能进行注释，这有助于理解基因的生物学功能和其在不同生物过程中的角色。

通过同源序列比对软件比对已知基因功能的数据库，将已识别的基因与其可能的生物学功能关联起来。

介绍内容：

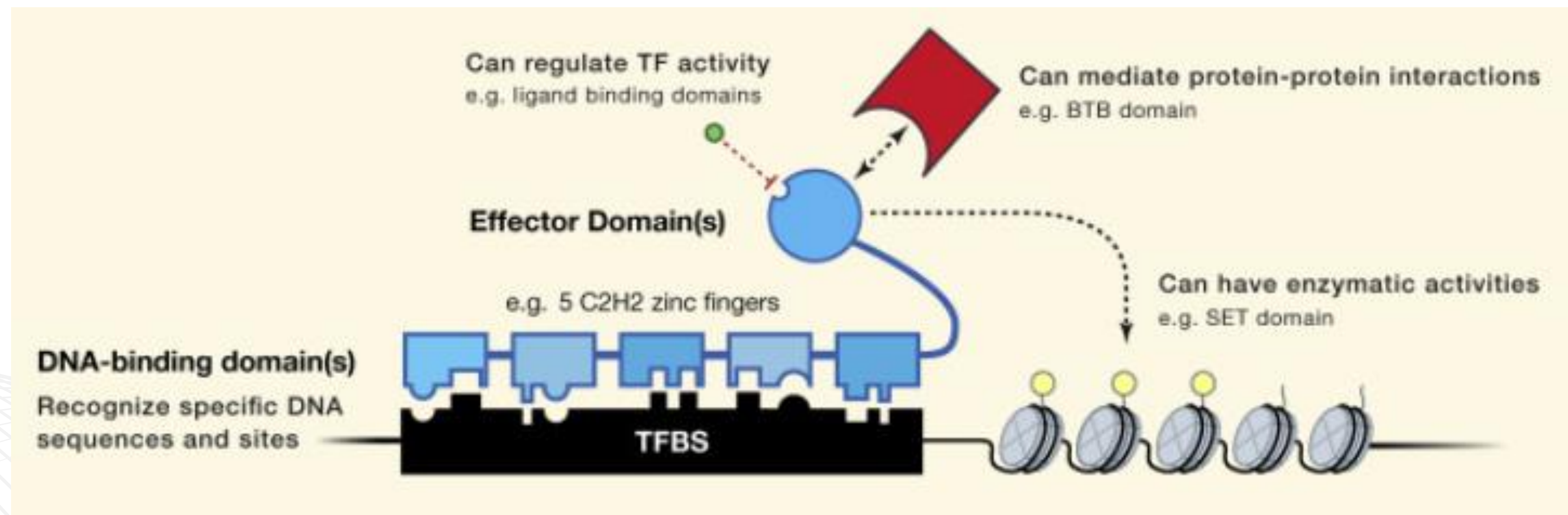
- ✓ 转录因子数据库注释
- ✓ 人类疾病类型注释
- ✓ 蛋白互作网络注释

转录因子注释

转录因子简介

转录因子是一类蛋白质，它们可以结合到DNA上，调控基因的转录，从而影响基因的表达。转录因子在生物体的发育、生长、细胞分化、免疫反应等许多生物过程中都起着关键作用。

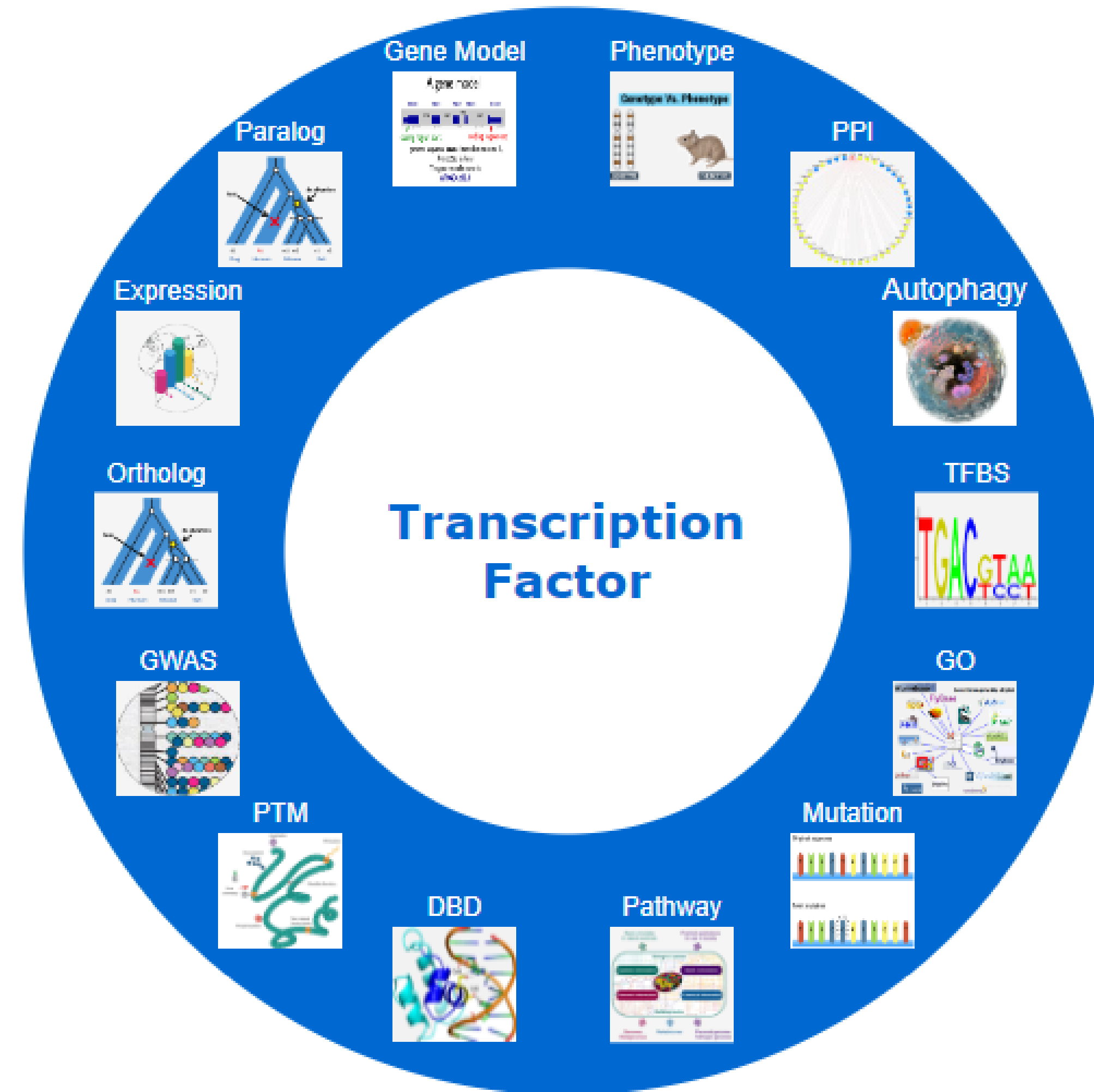
转录因子的研究是分子生物学和基因调控研究的重要部分，也是许多疾病，特别是癌症的研究的重要内容。



AnimalTFDB简介

AnimalTFDB是一个专门收集和整理动物转录因子的数据库。该数据库包含了大量的动物转录因子数据，包括转录因子的序列、结构、功能等信息。

<https://guolab.wchscu.cn/AnimalTFDB4//#/>



结果展示



cluster	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	GeneName	GeneID	TF_Family
0	4.99358170369067e-67	1.043186	0.844	0.302	1.27336333444112e-62	TBX18	ENSG00000112837	T-box
0	4.89215959044027e-66	1.436416	0.847	0.349	1.24750069556227e-61	ZNF331	ENSG00000130844	zf-C2H2
1	2.25074372048498e-08	0.345525	0.121	0.316	0.000574	SMARCC1	ENSG00000173473	MYB
2	3.62494038222404e-10	0.488692	0.885	0.721	9.24359797467129e-06	ZBTB16	ENSG00000109906	ZBTB

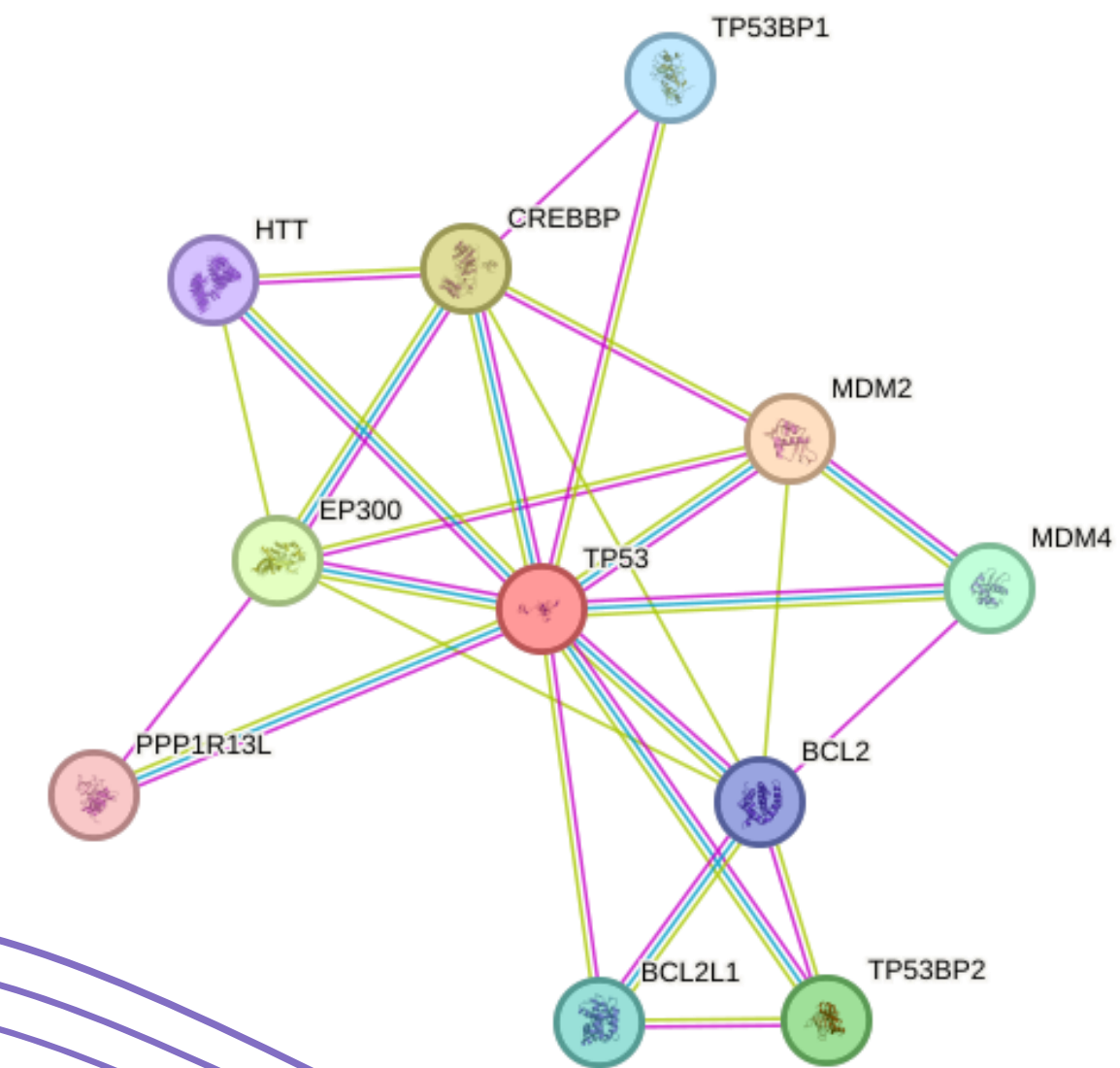
蛋白互作网络（PPI）注释

背景介绍

蛋白质互作网络（Protein-Protein Interaction Network, 简称 PPI Network）是描述蛋白质之间物理接触和功能关联的模型。在这个网络中，节点代表蛋白质，边代表蛋白质之间的互作关系。

蛋白质互作网络是由蛋白通过彼此之间的相互作用构成，来参与生物信号传递、基因表达调节、能量和物质代谢及细胞周期调控等生命过程的各个环节。

- 规模大
- 复杂性
- 动态性



PPI 应用简介



蛋白互作网络的应用:

疾病基因的发现

疾病机制的理解

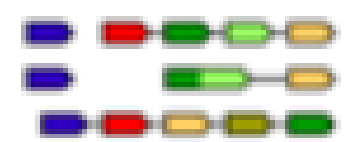
药物靶点的发现

药物副作用的预测

.....

STRING数据库介绍

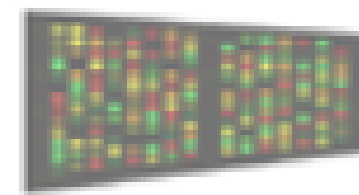
STRING数据库是一个搜寻已知蛋白质之间和预测蛋白质之间相互作用的系统，这种相互作用既包括蛋白质之间直接的物理的相互作用，也包括蛋白质之间间接的功能的相关性。
<https://string-db.org/>



Genomic Context
Predictions



High-throughput Lab
Experiments



(Conserved) Co-
Expression

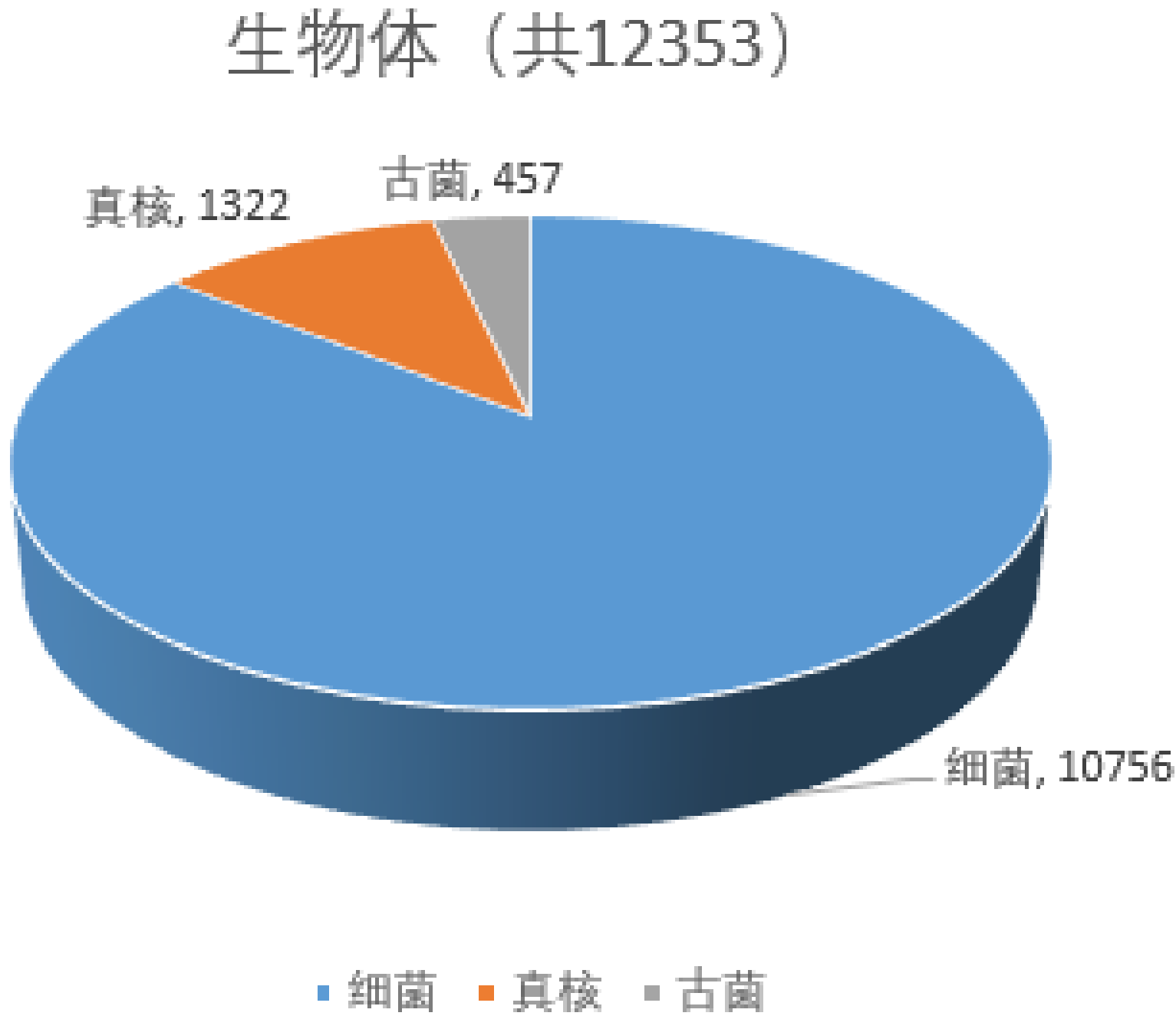


Automated
Textmining



Previous Knowledge in
Databases

STRING介绍



59,309,604 个蛋白

332,075,812个互作 (score>=0.9)
977,339,418个互作 (score>=0.7)

差异基因蛋白互作网络提取/注释

- 针对于数据库中包含的物种，直接从数据库中提取出目标基因集(比如差异基因list)的互作关系来构建网络；
- 针对于数据库中不包含的物种，我们首先将目标基因集中的序列应用blastx比对到string数据库中包含的参考物种的蛋白质序列上，并利用比对上的该参考物种的蛋白质互作关系构建互作网络。

差异基因蛋白互作网络



protein1	protein2	Combined_score	mode	action
ENSP000000000442	ENSP000000343925	900	binding	
ENSP000000000442	ENSP000000355865	567	binding	
ENSP000000003084	ENSP000000265563	900	binding	
ENSP000000003084	ENSP000000287878	800	inhibition	inhibition

Combined_score: 不同来源渠道的score值的组合，并进行了矫正

Mode: 相互作用类型 (reaction, expression, activation, ptmod(post-translational modifications), binding, catalysis)

Action: inhibition 或者activation

人类疾病类型注释

背景介绍

疾病与基因的关系是生物医学研究的重要领域。基因是生物体的基本遗传单位，它们编码生物体的所有蛋白质。这些蛋白质执行各种生物功能，包括细胞的生长、分化、代谢等。当这些基因发生突变，可能会导致蛋白质的功能发生改变，从而引发疾病。

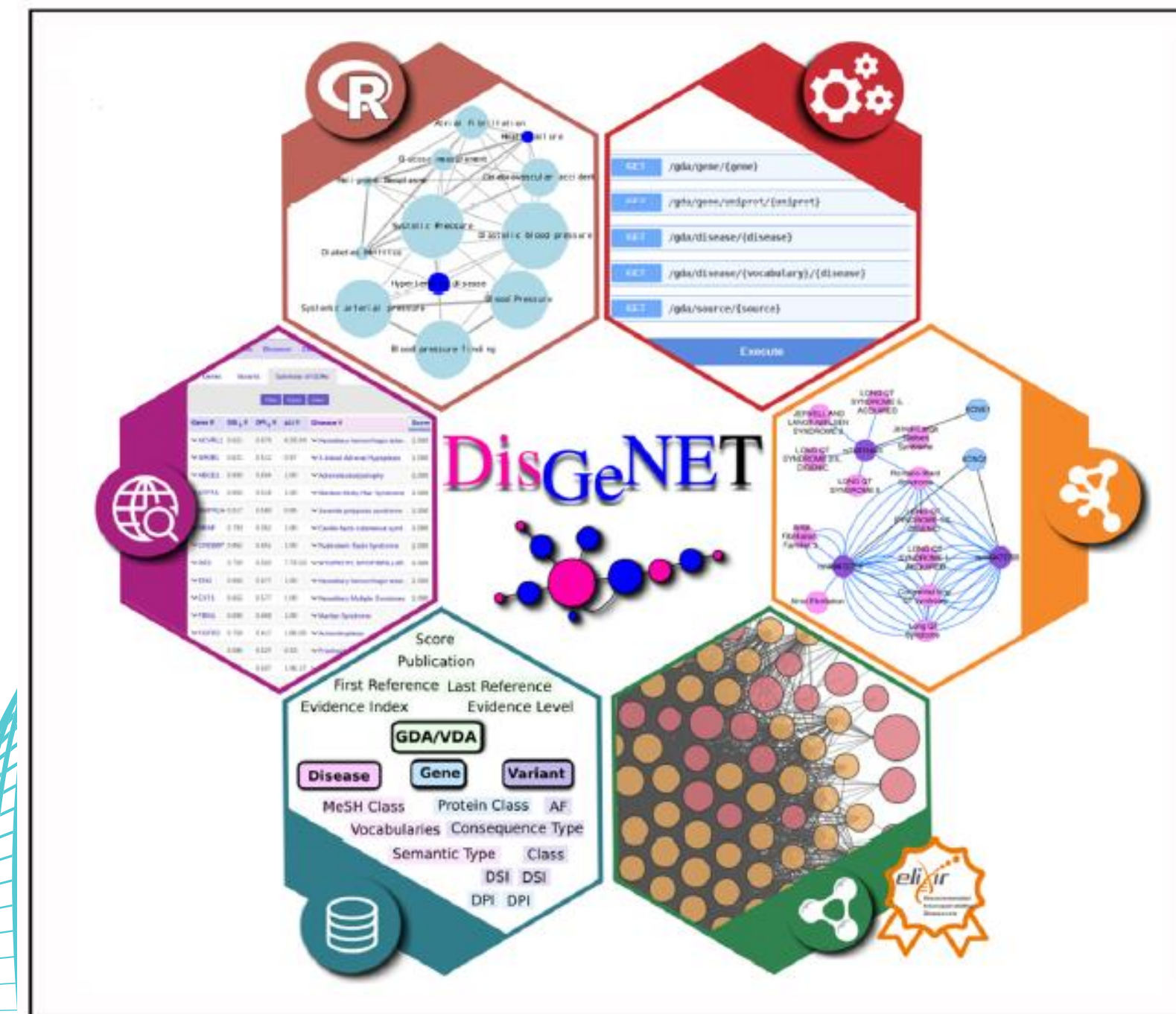
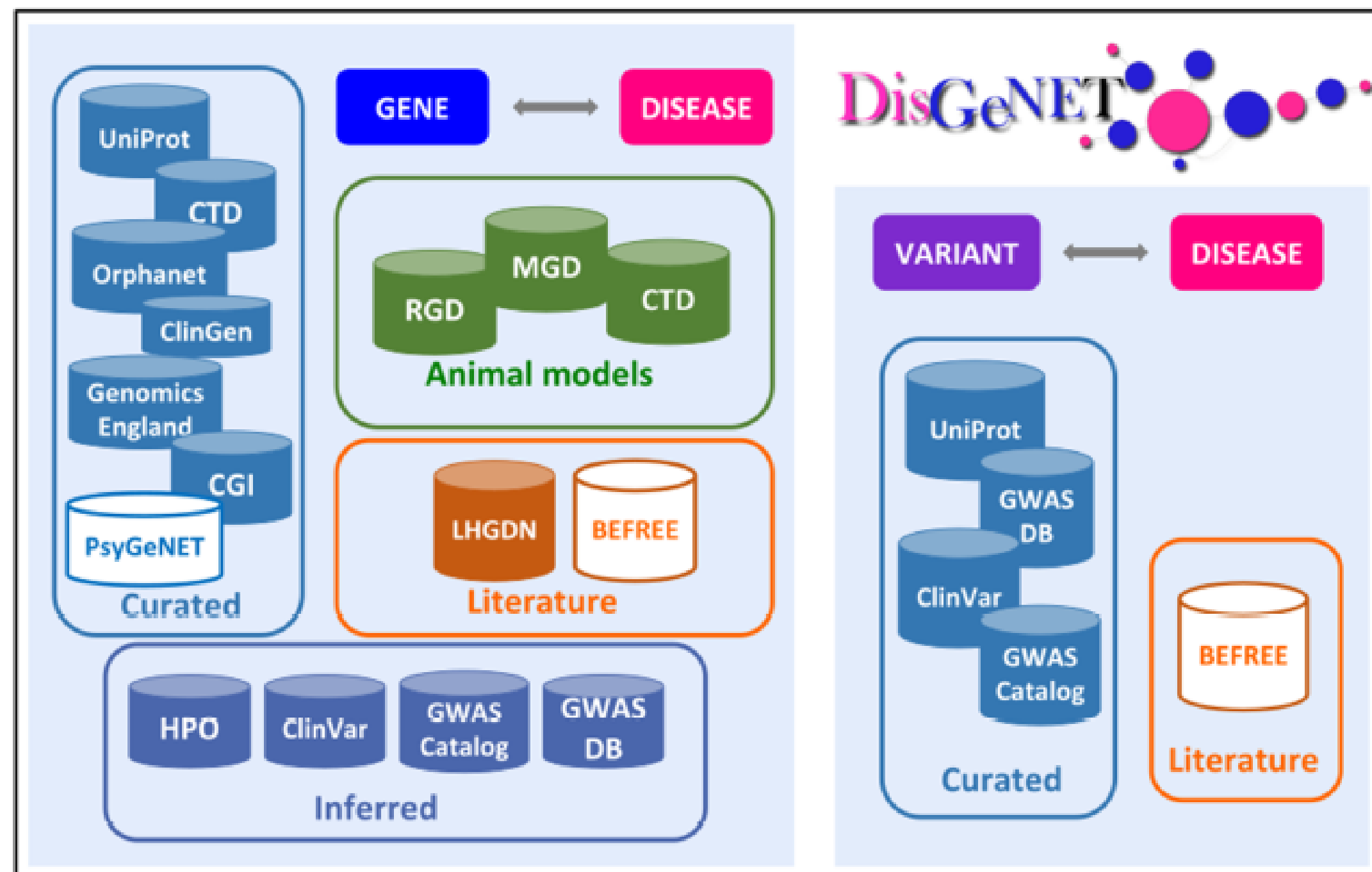
通过研究疾病与基因的关系，科研人员可以更好地理解疾病的发生机制，发现新的疾病标记物，开发新的治疗方法。例如，基因治疗就是一种直接针对疾病相关基因的治疗方法，通过修复或替换突变基因，来治疗疾病。

DisGeNET数据库介绍

DisGeNET是一个公开的、全面的、权威的疾病-基因关联数据库。它集成了多个数据源，包括基因组宽关联研究(GWAS)、动物模型、文献挖掘等，提供了大量的疾病-基因关联数据。

<https://www.disgenet.org/>

<https://www.disgenet.com/>



DisGeNET使用例子

- 阿尔兹海默症 (Alzheimer's disease) 相关的基因有哪些
- 哪些疾病与APP基因相关 (练习)
- 哪些疾病与转录因子变化有关

https://www.disgenet.org/browser/0/0/3/0/0/25/source__ALL/_b./

批量注释示例

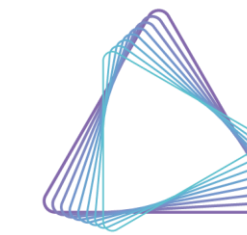


cluster	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	Gene Name	GeneID	Disease ID	Disease_Name	Disease Type	Disease_Class	Disease_Class_Name
0	4.99e-67	1.043186	0.844	0.302	1.27e-62	RBFOX1	ENSG00000078328	C0004352	Aultistic Disorder	disease	F03	Mental Disorders
0	4.89e-66	1.436416	0.847	0.349	1.24e-61	RBFOX1	ENSG00000078328	C0036572	Seizures	phenotype	C23;C10	Pathological Conditions, Signs and Symptoms; Nervous System Diseases
0	2.25e-08	0.345525	0.121	0.316	0.000574	RBFOX1	ENSG00000078328	C0022333	Jacksonian Seizure	disease	C23;C10	Pathological Conditions, Signs and Symptoms; Nervous System Diseases

富集疾病筛选

Disease ID	Disease_Name	DEGs with disease annotation (49)	All genes with disease annotation (9703)	RichFactor	Pvalue	Qvalue	Genelist
C1861678	Charcot-Marie-Tooth Disease, Axonal, Type 2a1	1(2.04%)	1(0.01%)	1	0	0	KIF1B,
C3150127	BARDET-BIEDL SYNDROME 15	1(2.04%)	1(0.01%)	1	0	0	WDPCP,
C0393519	Cerebellar Ataxia, Early Onset	1(2.04%)	1(0.01%)	1	0	0	SIL1,

应用介绍



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Identifying key genes of classic papillary thyroid cancer in women aged more than 55 years old using bioinformatics analysis

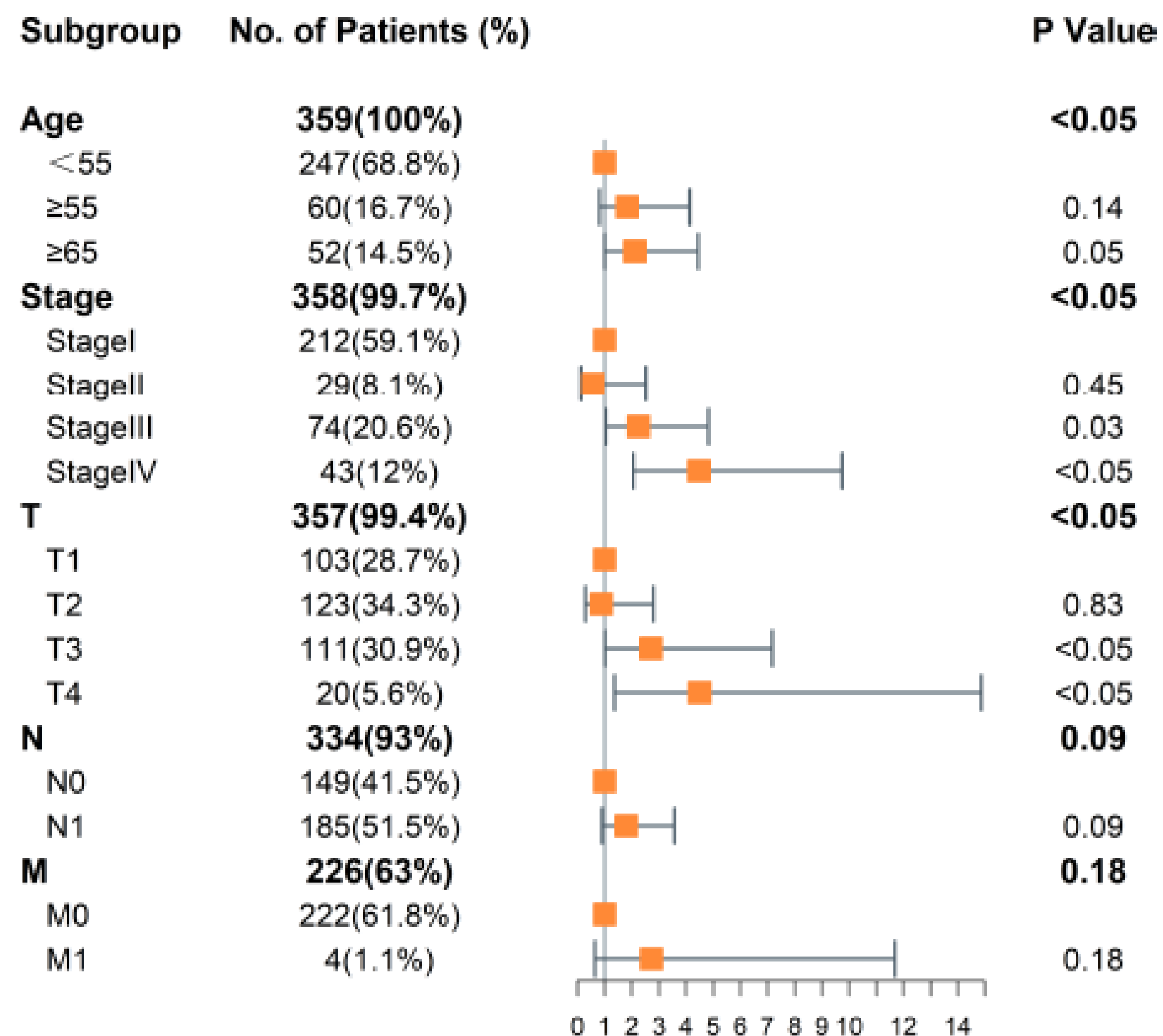
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Hazard Ratio Plot



7个低风险组，
11个高风险组

差异基因识别

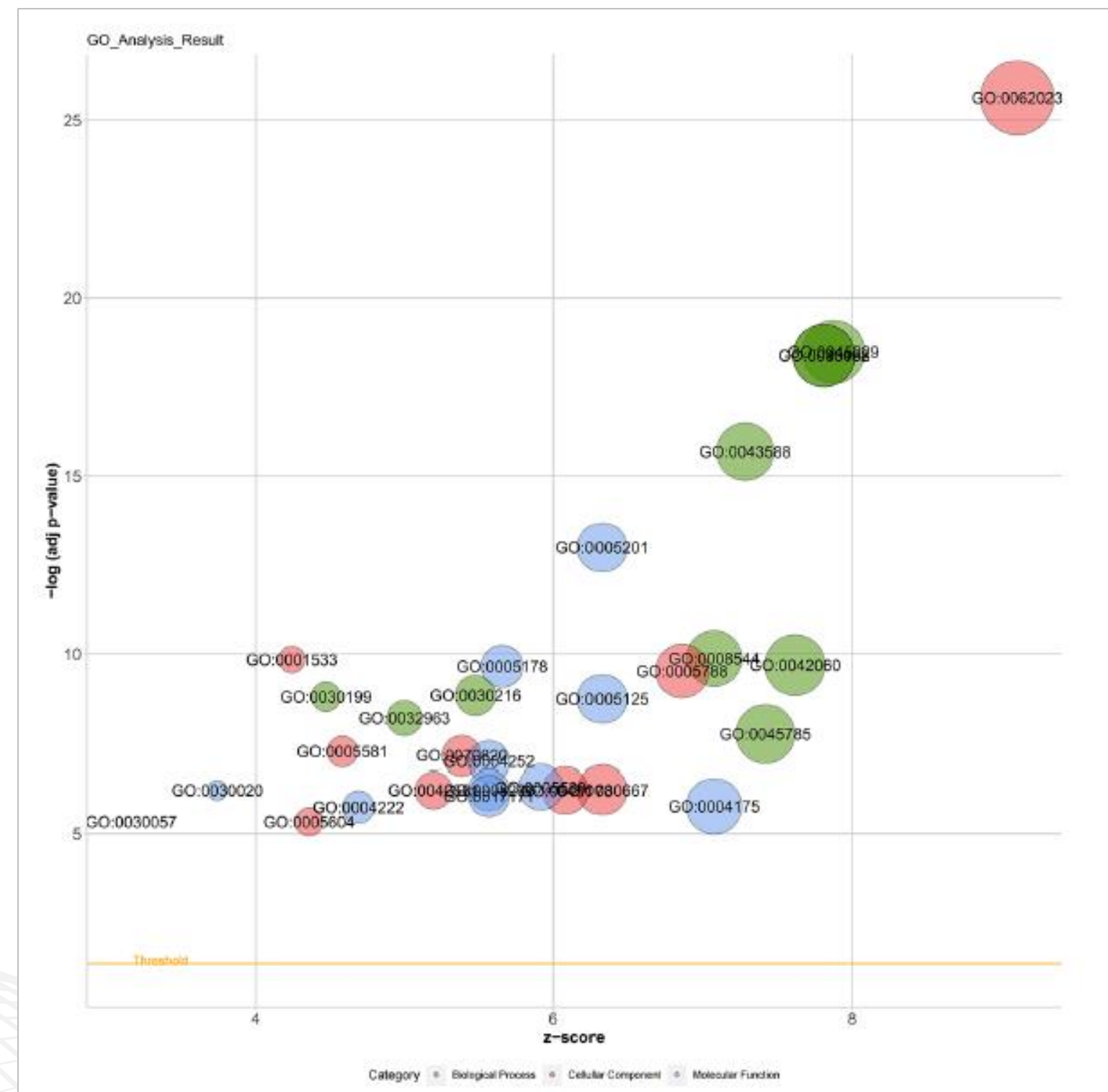
GO和KEGG富集分析

PPI网络构建

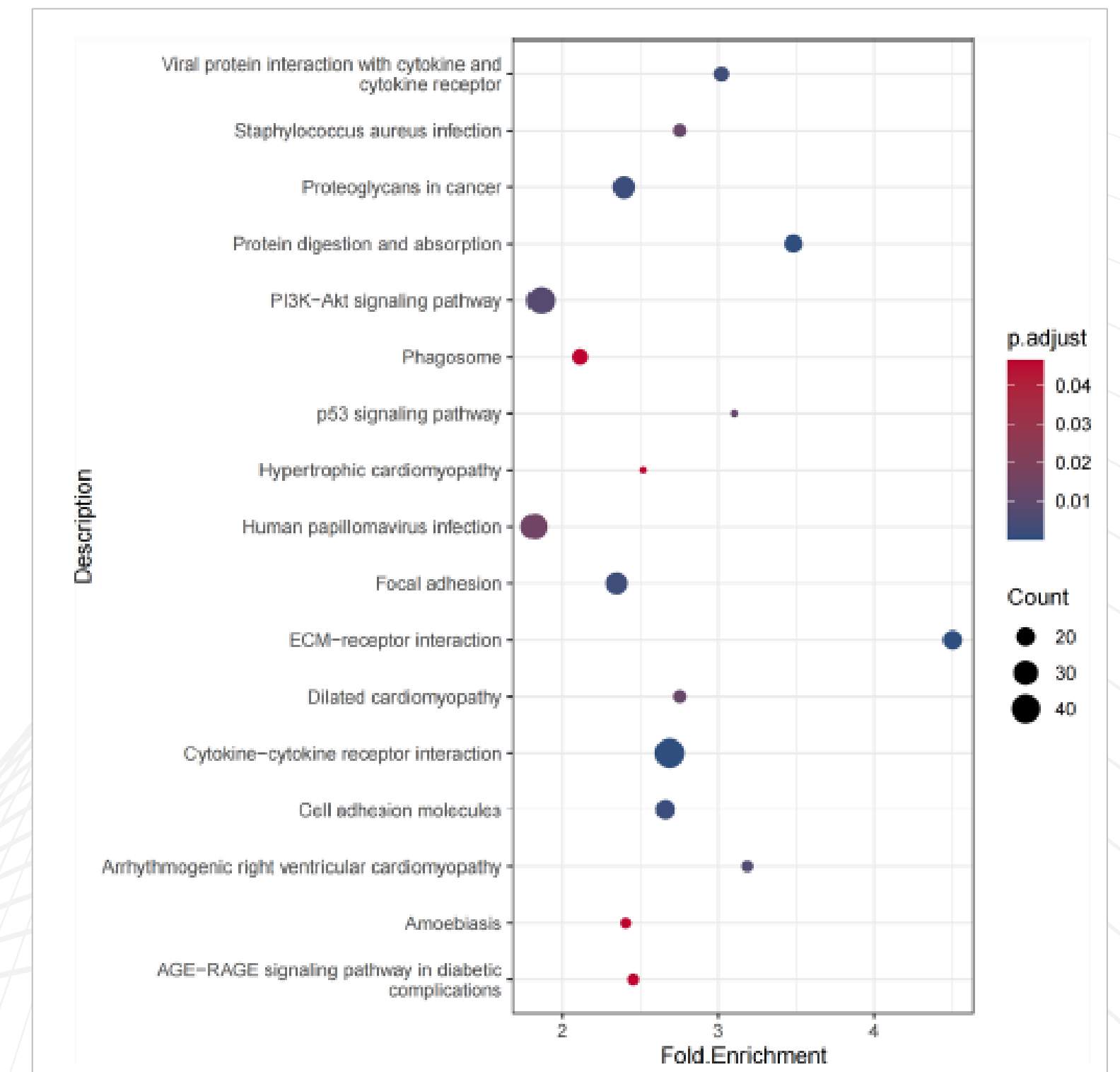
生存分析

TF和miRNA调控网络

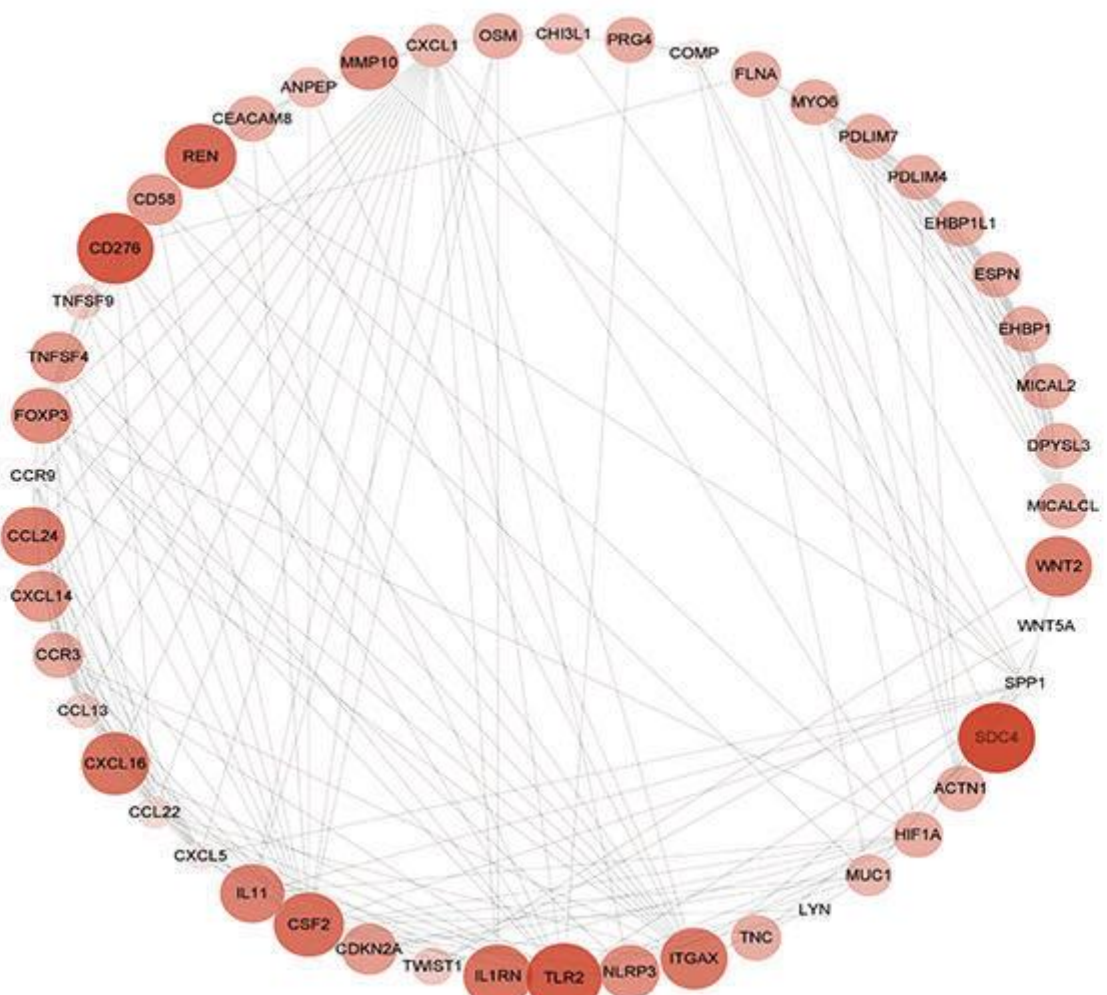
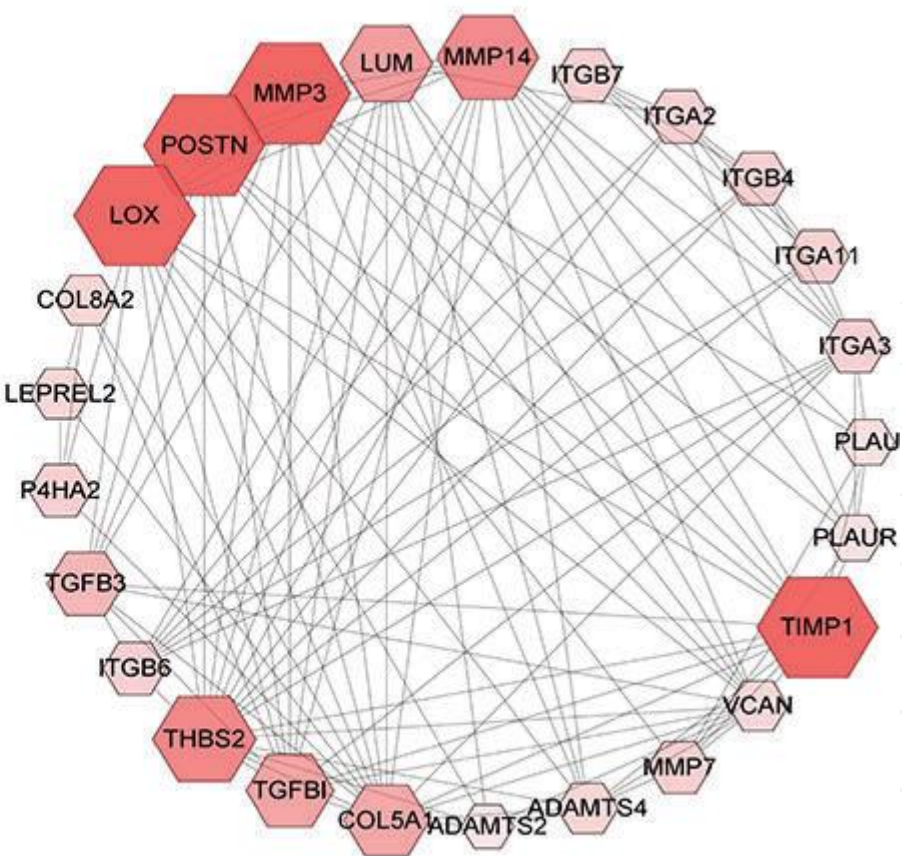
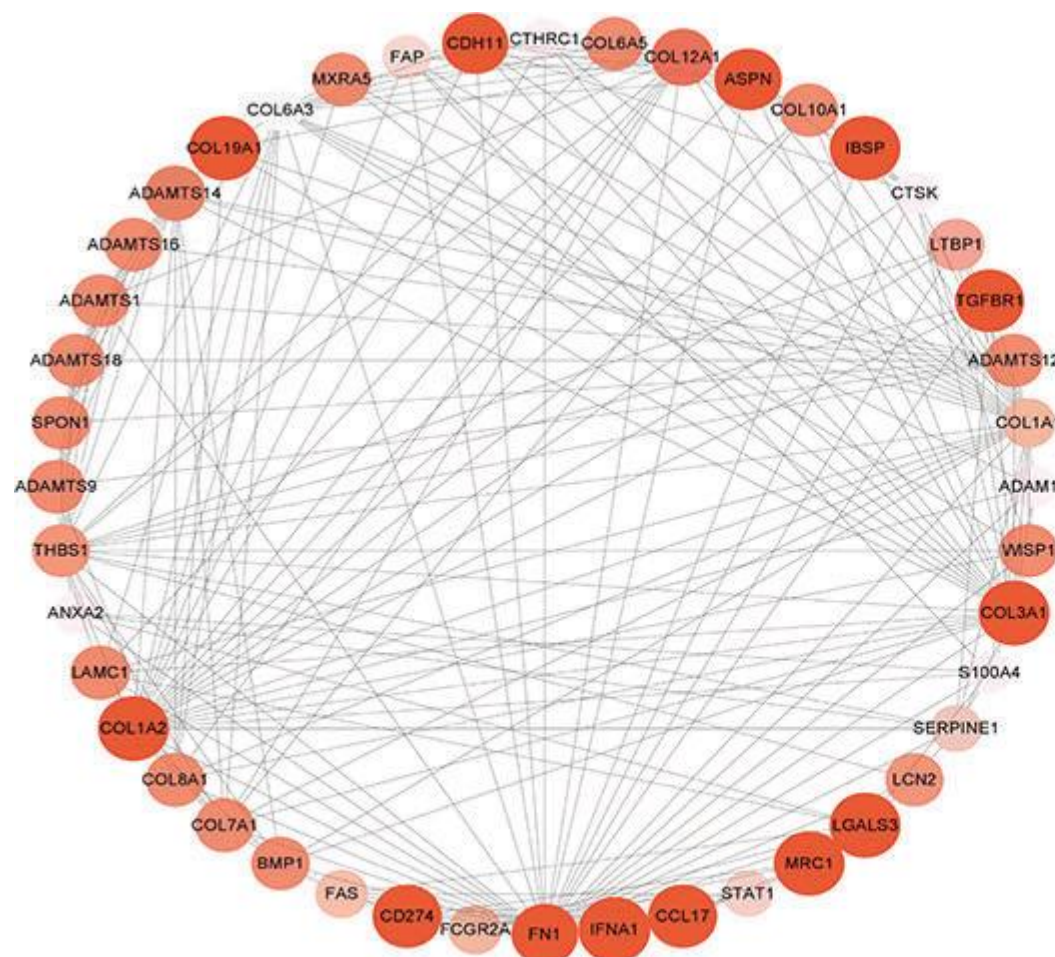
GO和Pathway富集分析



ID	Description
GO:0045229	external encapsulating structure organization
GO:0030198	extracellular matrix organization
GO:0043062	extracellular structure organization
GO:0043588	skin development
GO:0008544	epidermis development
GO:0042060	wound healing
GO:0030216	keratinocyte differentiation
GO:0030199	collagen fibril organization
GO:0032963	collagen metabolic process
GO:0045785	positive regulation of cell adhesion
GO:0062023	collagen-containing extracellular matrix
GO:0001533	cornified envelope
GO:0005788	endoplasmic reticulum lumen
GO:0005581	collagen trimer
GO:0070820	tertiary granule
GO:0042581	specific granule
GO:0030667	secretory granule membrane
GO:0045178	basal part of cell
GO:0030057	desmosome
GO:0005604	basement membrane
GO:0005201	extracellular matrix structural constituent
GO:0005178	integrin binding
GO:0005125	cytokine activity
GO:0004252	serine-type endopeptidase activity
GO:0005539	glycosaminoglycan binding
GO:0030020	extracellular matrix structural constituent conferring tensile strength
GO:0008236	serine-type peptidase activity
GO:0017171	serine hydrolase activity
GO:0004175	endopeptidase activity
GO:0004222	metalloendopeptidase activity

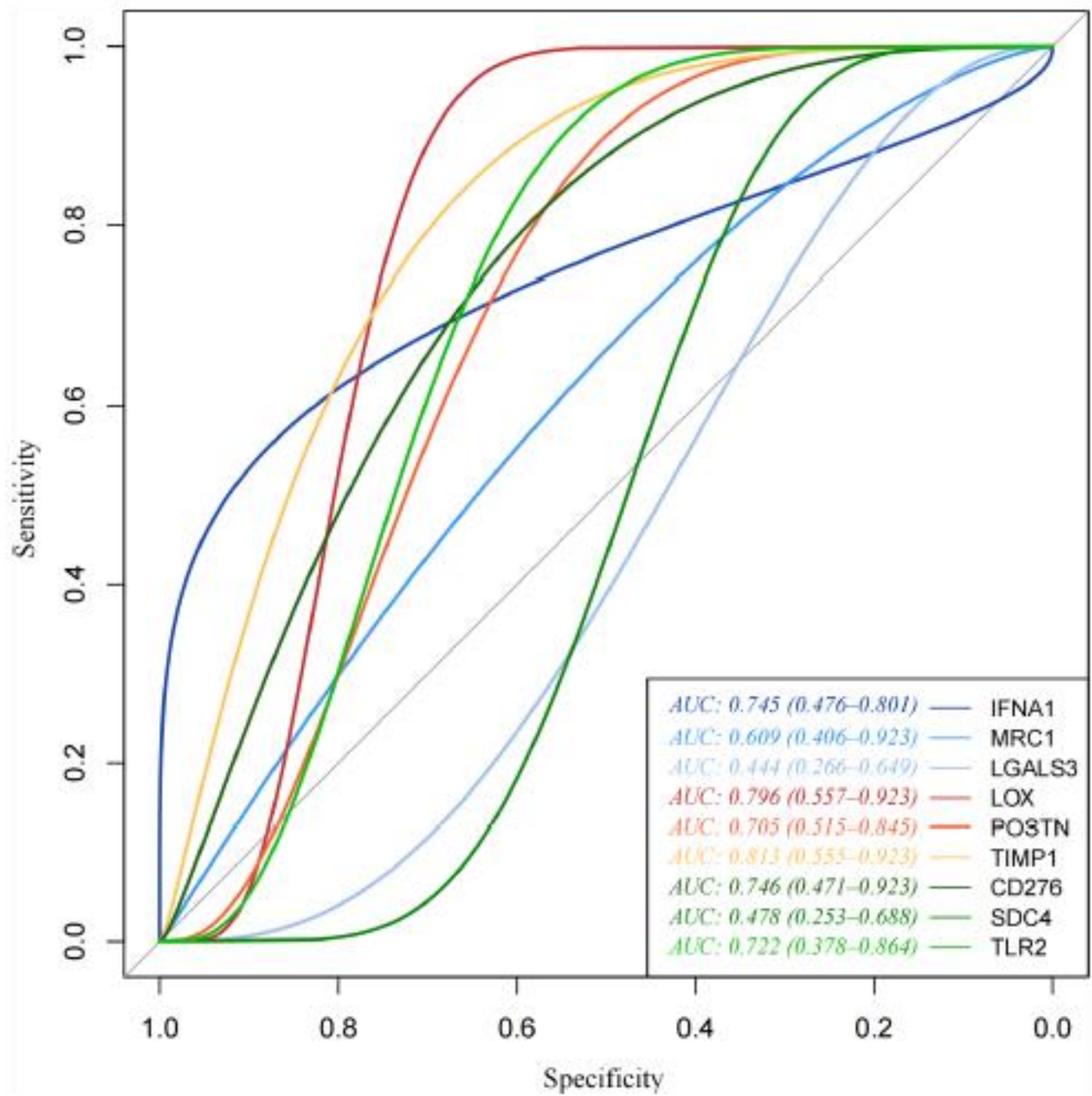


蛋白互作网络

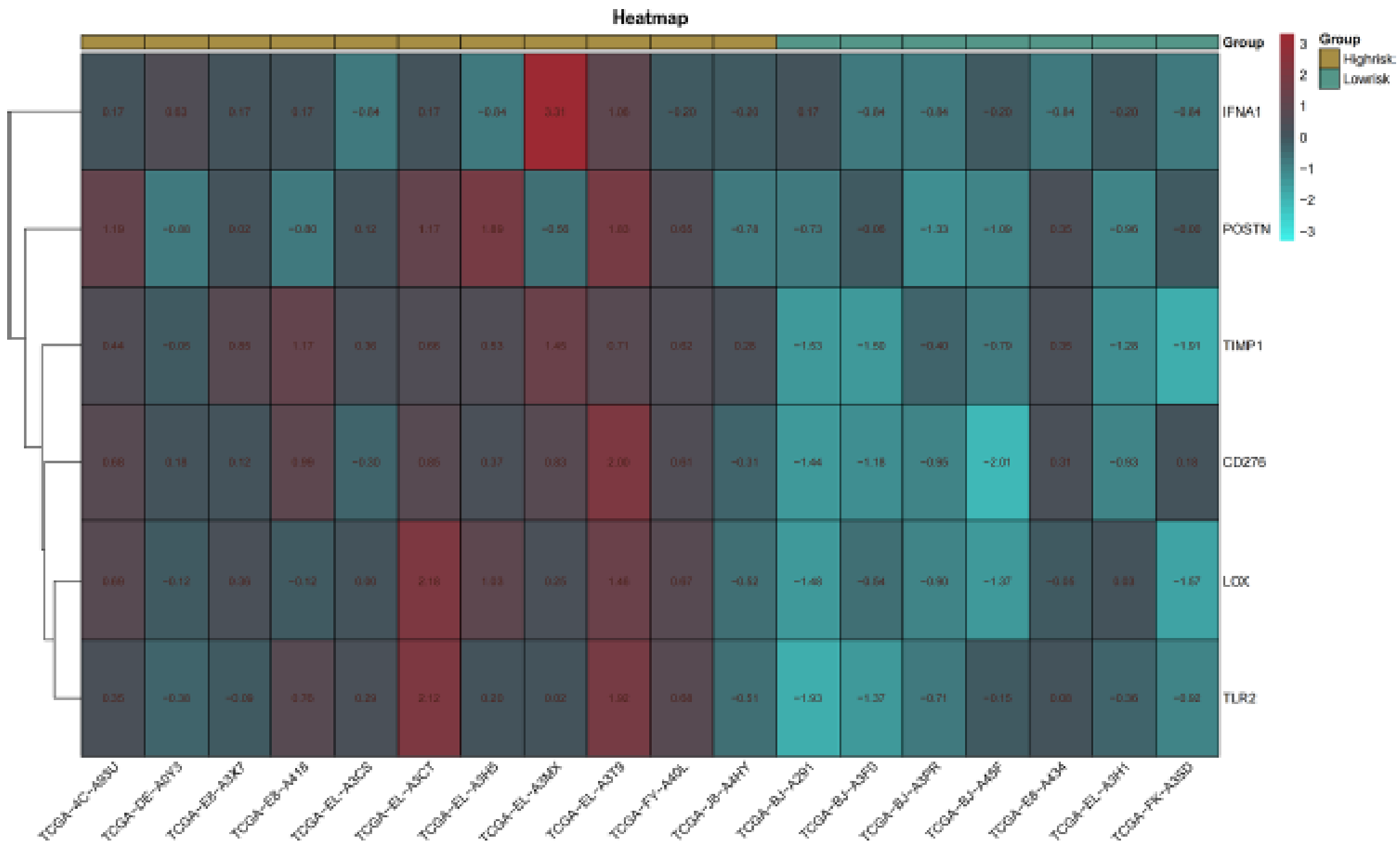


IFNA1, MRC1, LGALS3,
LOX, POSTN, TIMP1,
CD276, SDC4, TLR2

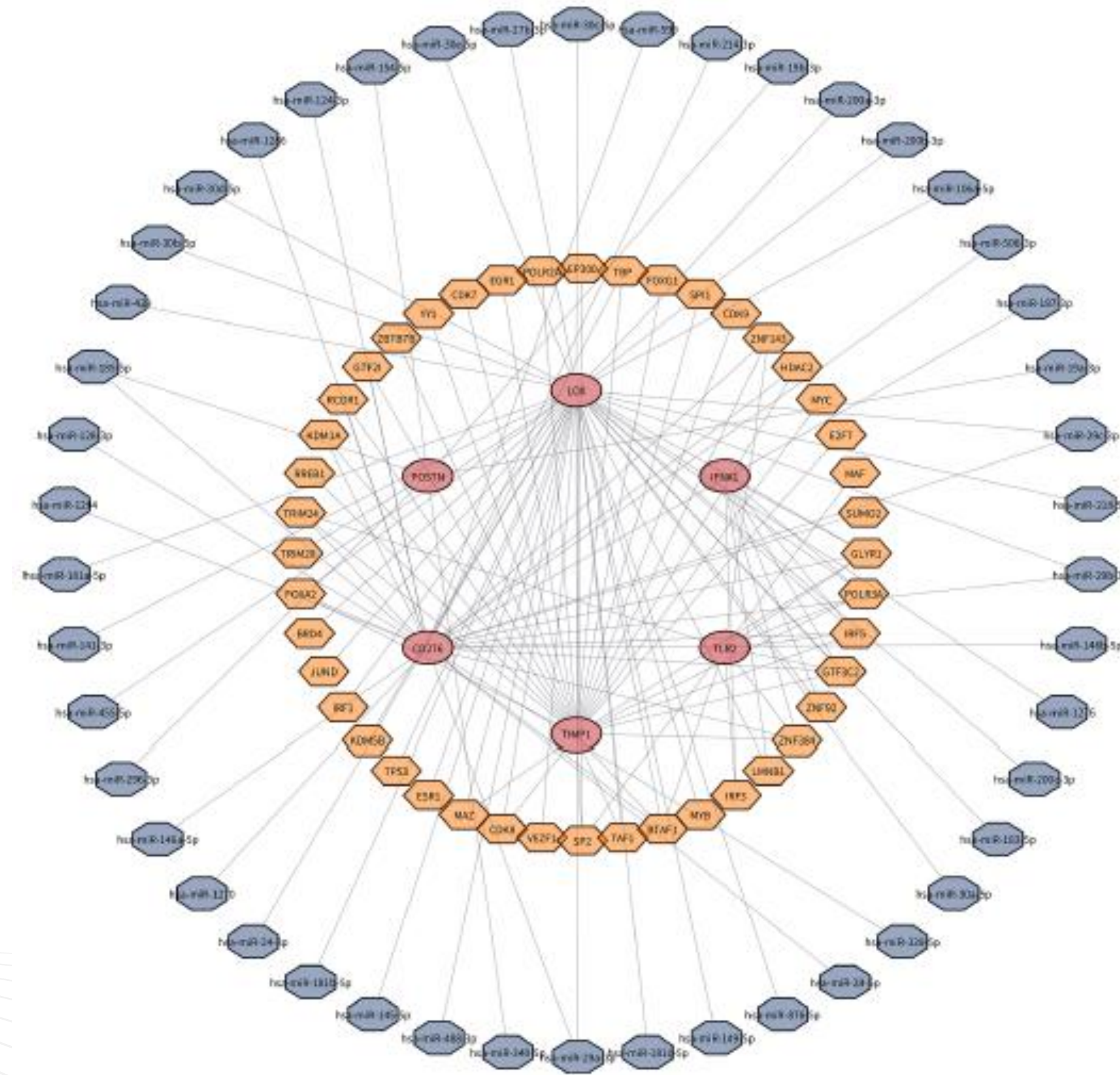
ROC分析



IFNA1, MRC1, LGALS3,
LOX, POSTN, TIMP1,
CD276, SDC4, TLR2

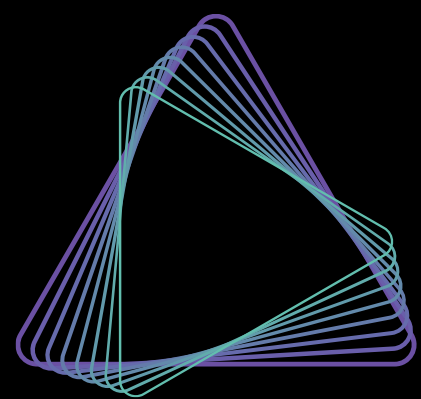


TF-miRNA-gene网络





Thank you!



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